Cloud Incident Monitoring Dashboard

Event-Driven Data Processing System — Deliverable Assignment

Project Title: Cloud Incident Monitoring Dashboard	**Architecture:** AWS Event-Driven Serverless
Deliverable Type: Assignment Documentation	**Status:** Completed & Submitted

1. Project Overview

The Cloud Incident Monitoring Dashboard is a real-time event-driven cloud application developed using AWS cloud services. The system automatically processes uploaded cloud incident data and displays it on a web dashboard without any manual intervention.

The application demonstrates an event-driven architecture where every uploaded incident is automatically processed using AWS Lambda, stored in Amazon DynamoDB, and displayed through an API on a dashboard hosted on Amazon EC2.

2. Objective

The primary objective of this project is to design and implement a scalable cloud-native solution capable of automatically processing cloud incident data using AWS services.

The system aims to:

- Automate cloud incident processing
- Eliminate manual data handling
- Provide real-time dashboard updates
- Visualize cloud infrastructure health
- Demonstrate serverless event-driven architecture

3. System Workflow

The complete workflow is as follows:

Step 1
The user uploads a JSON file containing cloud incident data into an Amazon S3 Bucket.

Step 2
Amazon S3 automatically detects the new object creation event.

Step 3
The Object Created Event invokes an AWS Lambda Function.

Step 4
The Lambda function validates, processes, and restructures the uploaded incident data.

Step 5
The processed data is stored inside Amazon DynamoDB.

Step 6
Amazon API Gateway exposes REST APIs for retrieving processed cloud incident data.

Step 7

The Cloud Incident Dashboard hosted on Amazon EC2 fetches the latest processed data through API Gateway and displays it dynamically.

4. AWS Services Used

Service	Purpose
Amazon EC2	Hosts the Cloud Dashboard
Amazon S3	Stores uploaded cloud incident files
AWS Lambda	Automatically processes uploaded data
Amazon DynamoDB	Stores processed incident information
Amazon API Gateway	Provides REST APIs
Amazon CloudWatch	Logs Lambda execution and monitors resources
AWS IAM	Manages secure permissions

5. Design Decisions

Several AWS services were selected to achieve a scalable and automated architecture:

- Amazon S3 was selected because it supports event notifications whenever new objects are uploaded.
- AWS Lambda was chosen to eliminate server management while automatically processing uploaded data.
- Amazon DynamoDB provides fast and scalable NoSQL storage for processed incidents.
- API Gateway allows secure communication between the backend and dashboard.
- Amazon EC2 hosts the web dashboard and makes it accessible through a browser.
- CloudWatch provides monitoring and logging for debugging and performance tracking.
- IAM ensures secure access between AWS services.

6. Challenges Faced

During development several challenges were encountered:

- Configuring IAM Roles and Policies
- Setting up S3 Event Notifications
- Lambda deployment and testing
- API Gateway CORS configuration
- Processing nested JSON data
- Integrating dashboard with REST APIs

These issues were resolved through AWS CloudWatch logs and proper IAM configuration.

7. Outcome

- The project successfully implements a fully automated cloud-native event-driven architecture.
- Whenever new cloud incident data is uploaded, the complete processing pipeline executes automatically without any manual intervention.

- The dashboard displays processed incident information in real time along with infrastructure statistics, analytics charts, and cloud monitoring metrics.

8. Conclusion

This project demonstrates the practical implementation of cloud-native architecture using AWS services. It successfully integrates Amazon S3, AWS Lambda, DynamoDB, API Gateway, CloudWatch, IAM, and EC2 into a scalable event-driven solution capable of processing and visualizing cloud incident data automatically.